

Burak Varıcı

Curriculum Vitae

Machine Learning Department
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Research Interests: I am broadly interested in the theoretical foundations of AI/ML, particularly in representation learning and causality. This involves developing novel theoretical results and scalable algorithms for identifiable representation learning, causal discovery, and intervention design, with a recent focus on applications in robotics, tabular data, and inverse problems.

Academic Positions

- 07.2026 – **University of Delaware**, Newark, DE.
 - present Assistant Professor, Electrical and Computer Engineering Department
 - Resident Faculty of the Data Science Institute
- 07.2024 – **Carnegie Mellon University**, Pittsburgh, PA.
 - 06.2026 ○ Postdoctoral Research Associate, Machine Learning Department
 - Host: Prof. Pradeep Ravikumar

Education

- 2024 **Rensselaer Polytechnic Institute**, Troy, NY.
 - Ph.D. in Electrical Computer, and Systems Engineering
 - Advisor: Prof. Ali Tajer
 - Dissertation: **Causal Learning via Interventions: Estimation and Design**
- 2020 **Rensselaer Polytechnic Institute**, Troy, NY.
 - M.S. in Electrical Computer, and Systems Engineering
- 2018 **Bogazici University**, Istanbul, Turkey.
 - B.S. in Electrical and Electronics Engineering

Professional Experience

- 09–12.2022 **MIT-IBM Watson AI Lab**, Cambridge, MA.
 - Visiting Research Scholar
 - Collaborators: Dr. Dmitriy A Katz, Dr. Prasanna Sattigeri, Dr. Dennis Wei
- 05–08.2020 **IBM Research**, remote.
 - Research Intern
 - Mentors: Dr. Karthikeyan Shanmugam, Dr. Prasanna Sattigeri
 - Project: Disentangled generative modeling
- 06–08.2017 **Speech Enabled Smart Technologies**, Istanbul, Turkey.
 - Research Intern
 - Project: Built neural networks for speaker identity verification.

Publications

* denotes equal contribution .

Selected Publications

- JMLR'25 **Score-based Causal Representation Learning: Linear and General Transformations**,
B. Varıcı, E. Acartürk, K. Shanmugam, A. Kumar, and A. Tajer.
Journal of Machine Learning Research, 2025
- JMLR'23 **Causal Bandits for Linear Structural Equation Models**,
B. Varıcı, K. Shanmugam, P. Sattigeri, and A. Tajer.
Journal of Machine Learning Research, 2023
- ICML'25 **Contextures: Representations from Contexts**,
R. Zhai, K. Yang, **B. Varıcı**, CP. Tsai, and P. Ravikumar.
International Conference on Machine Learning, 2025

Peer-reviewed Journals

- JMLR'25 **Score-based Causal Representation Learning: Linear and General Transformations**,
B. Varıcı, E. Acartürk, K. Shanmugam, A. Kumar, and A. Tajer.
Journal of Machine Learning Research, 2025
- TMLR'24 **Separability Analysis for Causal Discovery in Mixture of DAGs**,
B. Varıcı, D. Katz-Rogozhnikov, D. Wei, P. Sattigeri, and A. Tajer.
Transactions on Machine Learning Research, 2024
- JSAIT'24 **Robust Causal Bandits for Linear Models**,
Z. Yan, A. Mukherjee, **B. Varıcı**, and A. Tajer.
IEEE Journal on Selected Areas in Information Theory, 2024
- JMLR'23 **Causal Bandits for Linear Structural Equation Models**,
B. Varıcı, K. Shanmugam, P. Sattigeri, and A. Tajer.
Journal of Machine Learning Research, 2023

Peer-reviewed Conferences

AI/ML conferences are peer-reviewed with low acceptance rates (20-30%) .

- NeurIPS'26 **Beyond Eigenfunctions: Divergence Principal Functions for Representation Learning**,
R. Ray, S. Dharod, **B. Varıcı**, N. Boffi, and P. Ravikumar.
Conference on Neural Information Processing Systems, 2026
- ICML'25 **Contextures: Representations from Contexts**,
R. Zhai, K. Yang, **B. Varıcı**, CP. Tsai, and P. Ravikumar.
International Conference on Machine Learning, 2025
- AISTATS'25 **On the Consistent Recovery of Joint Distributions from Conditionals**,
M. Majid, R. Pukdee, V. Agrawal, **B. Varıcı**, and P. Ravikumar.
International Conference on Artificial Intelligence and Statistics, 2025
- NeurIPS'24 **Linear Causal Representation Learning from Unknown Multi-node Interventions**,
B. Varıcı, E. Acartürk, K. Shanmugam, and A. Tajer.
Conference on Neural Information Processing Systems, 2024
- NeurIPS'24 **Sample Complexity of Interventional Causal Representation Learning**,
E. Acartürk, **B. Varıcı**, K. Shanmugam, and A. Tajer.
Conference on Neural Information Processing Systems, 2024
- NeurIPS'24 **Interventional Causal Discovery in a Mixture of DAGs**,
B. Varıcı, D. Katz-Rogozhnikov, D. Wei, P. Sattigeri, and A. Tajer.
Conference on Neural Information Processing Systems, 2024
- AISTATS'24 **General Identifiability and Achievability for Causal Representation Learning**,
(oral) **B. Varıcı**, E. Acartürk, K. Shanmugam, and A. Tajer.
International Conference on Artificial Intelligence and Statistics, May 2024
- ISIT'24 **Improved Bound for Robust Causal Bandits with Linear Models**,
Z. Yan, A. Mukherjee, **B. Varıcı**, and A. Tajer.
International Symposium on Information Theory, 2024
- UAI'22 **Intervention Target Estimation in the Presence of Latent Variables**,
B. Varıcı, K. Shanmugam, P. Sattigeri, and A. Tajer.
The Conference on Uncertainty in Artificial Intelligence, 2022
- NeurIPS'21 **Scalable Intervention Target Estimation in Linear Models**,
B. Varıcı, K. Shanmugam, P. Sattigeri, and A. Tajer.
Conference on Neural Information Processing Systems, 2021
- AISTATS'21 **Learning Shared Subgraphs in Ising Model Pairs**,
B. Varıcı*, S. Sihag*, and A. Tajer.
International Conference on Artificial Intelligence and Statistics, 2021

Preprints and Others

- 2026 **Learning Generative Factors via Causal Representation Learning: A Score-based Perspective**,
E. Acartürk, **B. Varici**, K. Shanmugam, and A. Tajer.
IEEE BITS the Information Theory Magazine
- 2026 **Tabular Numeric Stretch Transformation**,
Z. Ye, J. Kim, J. Sundberg, **B. Varici**, and P. Ravikumar.
arXiv:2608.09162, 2026
- 2025 **Eigenfunction Extraction for Ordered Representation Learning**,
B. Varici*, C.P. Tsai*, R. Ray, N. Boffi, and P. Ravikumar.
arXiv:2510.24672
- 2025 **ROPES: Robotic Pose Estimation via Score-based Causal Representation Learning**,
P. Kulkarni, P. Datta, E. Acartürk, **B. Varici**, K. Shanmugam, and A. Tajer.
arXiv:2510.20884, *NeurIPS 2025 Workshop on Embodied World Models*
- 2023 **Score-based Causal Representation Learning with Interventions**,
B. Varici, E. Acartürk, K. Shanmugam, A. Kumar, and A. Tajer.
arXiv:2301.08230, 2023

Book Chapters

- 2025 Causal Representation Learning,
B. Varici*, C. Squires*, and P. Ravikumar.
Book Chapter in **Neuro-symbolic AI: Foundations and Applications**, Wiley, April 2026.

Honors & Awards

- 2025 IEEE Signal Processing Society Best PhD Dissertation Award
- 2025 NeurIPS Top Reviewer
- 2020–2024 IBM AI Horizons Fellowship
- 2024 Allen B. Dumont Prize (RPI ECSE doctoral dissertation award)
- 2023 NeurIPS Top Reviewer, UAI Top Reviewer
- 2022 Belsky Award for RPI Computational Sciences and Engineering
- 2022 Jerry Dziuba ECSE Graduate Student Service Award
- 2013–2018 TUBITAK (Turkish Equivalent of NSF) Undergraduate Fellowship
- 2012 Silver Medal, Turkish National Mathematical Olympiad
- 2012 Silver Medal, International Balkan Mathematical Olympiad
- 2010 Gold Medal, International Junior Balkan Mathematical Olympiad

Talks

Identifiable Representation Learning

- 01.2026 **INFORMED-AI Seminar Series - University of Bristol**

Causal Representation Learning

- 10.2025 **CMU - Statistics and Machine Learning Reading Group**
- 10.2025 **KUIS (Koc University & Is Bank) AI Center**
- 03.2025 **Artificial Intelligence with Causal Techniques Workshop at AAAI**
- 02.2025 **Booth School of Business at University of Chicago – Aragam Group**
- 03.2024 **CMU - Statistical & Symbolic Learning Group**
- 12.2023 **NeurIPS Causal Representation Learning Workshop**

Causal Bandits

- 02.2023 **IBM Causal Reinforcement Learning Group**

Teaching

- Fall 2026 **Instructor**, University of Delaware.
- **ELEG 467/667: Topics in Causality and Machine Learning**
Developed and taught a new course on foundations of modern causal inference and connections to machine learning, including an introduction to areas of active research such as representation learning.

- 2024–2025 **Guest Lecturer**, Carnegie Mellon University.
- CMU 10716: Advanced Machine Learning - Clustering
 - CMU 10716: Advanced Machine Learning - Causality
 - CMU 10741: Representation Learning - Causal Representation Learning
- 2018–2020 **Teaching Assistant**, Rensselaer Polytechnic Institute.
- ECSE 2410: Signals and Systems
 - ECSE 2610: Computer Components and Operations
 - ECSE 1010: Introduction to Electrical, Component, and Systems Engineering

Mentorship

- 2025–2026 Utkarsh Priyam (CMU M.S. → CMU Ph.D.)
- 2025–2026 Zihao Ye (CMU M.S. → Woodwide AI)
- 2025 Arnav Mantro (CMU Undergrad → Bloomberg Data Science)
- 2025 Hugo Contant (CMU Undergrad)
- 2024 Vishwajeet Agrawal (CMU M.S. → Skild AI)
- 2024 Mahbod Majod (CMU Ph.D. → MIT Ph.D.)
- 2022–2024 Emre Acartürk (RPI Ph.D.)
- 2022–2024 Zirui Yan (RPI Ph.D.)

Academic Service

- Organizer Co-organizer of **Identifiable Representation Learning Workshop** in October 2026, The Banff International Research Station, Banff, Alberta, Canada
- Tutorial Co-lead presenter of Causal Representation Learning Tutorial, **AAAI 2025**, Philadelphia, PA
- Tutorial (Contributor) Causal Representation Learning at **NASIT 2025** and **ICASSP 2026**.
- Journal Reviewer for IEEE Transactions on Signal Processing, IEEE Transactions on Pattern Recognition and Machine Intelligence, Transactions on Machine Learning Research
- Conference Regularly reviewing for conferences NeurIPS, ICML, AISTATS, AAAI, UAI, CLeaR since 2021
- Committee Secretary at RPI ECSE Graduate Student Council (2019–2022)
- Member